PRAKTIKUM BASIS DATA

MODUL 8

Scripting SQL (Create index, Create view, Analyze Table)

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

MODUL 8

TUGAS JURNAL

Kerjakan studi kasus pada modul 8 (halaman 87). Sertakan rekaman layar untuk setiap nomor.

8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
   b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
   c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

Jawaban:

Oracle SQL Developer : JURNAL_Modul08_Basdat_103122430009

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Connections

Oracle Connections
- DB_FILM
- DB_FILM_Jurnal06_Basdat_103122430009
- DB_FILM_Modul04
- DB_Film_modul05_Basdat_103122430009
- JURNAL_Modul02_Basdat_103122430009
- JURNAL_Modul08_Basdat_103122430009
- modul03_Basdat_Semester 4 SE08
- Modul05_Basdat_103122430009
- Modul08_Basdat_103122430009
- TP_Modul03_Basdat_103122430009
- TP_Modul04_Basdat_103122430009
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- Data Dictionary Reports
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- OLAP Reports
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- XML

Modul08_Basdat_103122430009.sql | JURNAL_Modul08_Basdat_103122430009

Worksheet | Query Builder

```
    last_name VARCHAR2(50) NOT NULL,
    email VARCHAR2(50),
    salary NUMBER,
    commission_pct NUMBER,
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_emp_dept FOREIGN KEY (department_id) REFERENCES department(department_id)
);

CREATE TABLE pemasok (
    pemasok_id NUMBER PRIMARY KEY,
    nama_pemasok VARCHAR2(100),
    kota VARCHAR2(50)
);

CREATE TABLE persediaan_barang (
    barang_id NUMBER PRIMARY KEY,
    nama_barang VARCHAR2(100),
    stok NUMBER,
    pemasok_id NUMBER,
    CONSTRAINT fk_brg_pemasok FOREIGN KEY (pemasok_id) REFERENCES pemasok(pemasok_id)
);
```

Script Output | Task completed in 0.05 seconds

```
Table DEPARTMENT created.

Table EMPLOYEE created.

Table PEMASOK created.

Table PERSEDIAAN_BARANG created.
```

Line 46 Column 3 | Insert | Modified | Windows: CR

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Oracle Connections
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- DB_FILM_Modul04
- DB_Film_modul05_Basdat_103122430009
- JURNAL_Modul02_Basdat_103122430009
- JURNAL_Modul08_Basdat_103122430009
- modul03_Basdat_Semester 4 SE08
- Modul05_Basdat_103122430009
- Modul08_Basdat_103122430009
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Modul08_Basdat_103122430009.sql | JURNAL_Modul08_Basdat_103122430009

Worksheet | Query Builder

```
    CONSTRAINT fk_emp_dept FOREIGN KEY (department_id) REFERENCES department(department_id)
);

CREATE TABLE pemasok (
    pemasok_id NUMBER PRIMARY KEY,
    nama_pemasok VARCHAR2(100),
    kota VARCHAR2(50)
);

CREATE TABLE persediaan_barang (
    barang_id NUMBER PRIMARY KEY,
    nama_barang VARCHAR2(100),
    stok NUMBER,
    pemasok_id NUMBER,
    CONSTRAINT fk_brg_pemasok FOREIGN KEY (pemasok_id) REFERENCES pemasok(pemasok_id)
);

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE pemasok COMPUTE STATISTICS;
ANALYZE TABLE persediaan_barang COMPUTE STATISTICS;
```

Script Output | Task completed in 0.073 seconds

```
Table EMPLOYEE created.

Table PEMASOK created.

Table PERSEDIAAN_BARANG created.

Table DEPARTMENT analyzed.

Table EMPLOYEE analyzed.

Table PEMASOK analyzed.

Table PERSEDIAAN_BARANG analyzed.
```

Line 51 Column 5 | Insert | Modified | Windows: CR

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Connections

Oracle Connections
- DB_FILM
- DB_FILM_Jurnal06_Basdat_103122430009
- DB_FILM_Modul04
- DB_Film_modul05_Basdat_103122430009
- JURNAL_Modul02_Basdat_103122430009
- JURNAL_Modul08_Basdat_103122430009
- modul03_Basdat_Semester 4 SE08
- Modul05_Basdat_103122430009
- Modul08_Basdat_103122430009
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Modul08_Basdat_103122430009.sql | JURNAL_Modul08_Basdat_103122430009

Worksheet | Query Builder

```
INSERT INTO employee VALUES (8, 'Scott', 'scott@mail.com', 9000, 0.25, SYSDATE, 80);
INSERT INTO employee VALUES (9, 'King', 'king@mail.com', 10000, NULL, SYSDATE, 80);
INSERT INTO employee VALUES (10, 'Turner', 'turner@mail.com', 6500, 0.1, SYSDATE, 90);

INSERT INTO pemasok VALUES (1, 'PT Maju Jaya', 'Jakarta');
INSERT INTO pemasok VALUES (2, 'CV Elektronik Bersama', 'Bandung');
INSERT INTO pemasok VALUES (3, 'PT Kertas Nasional', 'Surabaya');
INSERT INTO pemasok VALUES (4, 'Toko Alat Tulis', 'Semarang');
INSERT INTO pemasok VALUES (5, 'PT Tech Indo', 'Jakarta');
INSERT INTO pemasok VALUES (6, 'Distributor Kayu', 'Yogyakarta');
INSERT INTO pemasok VALUES (7, 'PT Plastik Berkah', 'Bekasi');
INSERT INTO pemasok VALUES (8, 'Grosir Komputer', 'Tangerang');
INSERT INTO pemasok VALUES (9, 'Pabrik Kaca', 'Bogor');
INSERT INTO pemasok VALUES (10, 'CV Besi Baja', 'Depok');

INSERT INTO persediaan_barang VALUES (1, 'Laptop', 50, 5);
INSERT INTO persediaan_barang VALUES (2, 'Kertas HVS', 200, 3);
INSERT INTO persediaan_barang VALUES (3, 'Meja Kantor', 20, 6);
INSERT INTO persediaan_barang VALUES (4, 'Kursi Kantor', 40, 7);
INSERT INTO persediaan_barang VALUES (5, 'Printer', 15, 2);
INSERT INTO persediaan_barang VALUES (6, 'Spidol', 100, 4);
INSERT INTO persediaan_barang VALUES (7, 'Monitor', 30, 8);
INSERT INTO persediaan_barang VALUES (8, 'Papan Tulis', 10, 9);
INSERT INTO persediaan_barang VALUES (9, 'Lemari Arsip', 25, 10);
INSERT INTO persediaan_barang VALUES (10, 'Mouse', 80, 5);
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.049 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	IT	101	1000
2	20	HR	102	1000
3	30	Finance	103	1001
4	40	Marketing	104	1001
5	50	Shipping	105	1002
6	60	Purchasing	106	1002
7	70	Public Relations	107	1003
8	80	Sales	108	1003
9	90	Executive	109	1004
10	100	Accounting	110	1004

Click on an identifier with the Control key down to perform "Go to Declaration"

| Line 102 Column 16 | Insert | Modified | Windows: C

Oracle SQL Developer : JURNAL_Modul08_Basdat_103122430009

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Connections

Oracle Connections
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- DB_FILM_Modul04
- DB_Film_modul05_Basdat_103122430009
- JURNAL_Modul02_Basdat_103122430009
- JURNAL_Modul08_Basdat_103122430009
- modul03_Basdat_Semester 4 SE08
- Modul05_Basdat_103122430009
- Modul08_Basdat_103122430009
- TP_Modul03_Basdat_103122430009
- TP_Modul04_Basdat_103122430009
- TP_Modul05_Basdat_103122430009
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Modul08_Basdat_103122430009.sql | JURNAL_Modul08_Basdat_103122430009

Worksheet | Query Builder

```
INSERT INTO pemasok VALUES (3, 'PT Kertas Nasional', 'Surabaya');
INSERT INTO pemasok VALUES (4, 'Toko Alat Tulis', 'Semarang');
INSERT INTO pemasok VALUES (5, 'PT Tech Indo', 'Jakarta');
INSERT INTO pemasok VALUES (6, 'Distributor Kayu', 'Yogyakarta');
INSERT INTO pemasok VALUES (7, 'PT Plastik Berkah', 'Bekasi');
INSERT INTO pemasok VALUES (8, 'Grosir Komputer', 'Tangerang');
INSERT INTO pemasok VALUES (9, 'Pabrik Kaca', 'Bogor');
INSERT INTO pemasok VALUES (10, 'CV Besi Baja', 'Depok');

INSERT INTO persediaan_barang VALUES (1, 'Laptop', 50, 5);
INSERT INTO persediaan_barang VALUES (2, 'Kertas HVS', 200, 3);
INSERT INTO persediaan_barang VALUES (3, 'Meja Kantor', 20, 6);
INSERT INTO persediaan_barang VALUES (4, 'Kursi Kantor', 40, 7);
INSERT INTO persediaan_barang VALUES (5, 'Printer', 15, 2);
INSERT INTO persediaan_barang VALUES (6, 'Spidol', 100, 4);
INSERT INTO persediaan_barang VALUES (7, 'Monitor', 30, 8);
INSERT INTO persediaan_barang VALUES (8, 'Papan Tulis', 10, 9);
INSERT INTO persediaan_barang VALUES (9, 'Lemari Arsip', 25, 10);
INSERT INTO persediaan_barang VALUES (10, 'Mouse', 80, 5);

COMMIT;
SET FEEDBACK ON;

SELECT * from department;
SELECT * from employee;
SELECT * from pemasok;
SELECT * from persediaan_barang;

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE pemasok COMPUTE STATISTICS;
ANALYZE TABLE persediaan_barang COMPUTE STATISTICS;
```

Script Output | Query Result

Task completed in 0.081 seconds

```
Table DEPARTMENT analyzed.


Table EMPLOYEE analyzed.


Table PEMASOK analyzed.


Table PERSEDIAAN_BARANG analyzed.
```

Click on an identifier with the Control key down to perform "Go to Declaration"

| Line 110 Column 3 | Insert | Modified | Windows: C

Oracle SQL Developer : JURNAL_Modul08_Basdat_103122430009

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Connections

Oracle Connections
- DB_FILM
- DB_FILM_Jurnal06_Basdat_103122430009
- DB_FILM_Modul04
- DB_Film_modul05_Basdat_103122430009
- JURNAL_Modul02_Basdat_103122430009
- JURNAL_Modul08_Basdat_103122430009
- modul03_Basdat_Semester 4 SE08
- Modul05_Basdat_103122430009
- Modul08_Basdat_103122430009
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JURNAL_Modul08_Basdat_103122430009

Worksheet | Query Builder

```
INSERT INTO persediaan_barang VALUES (1, 'Laptop', 50, 5);
INSERT INTO persediaan_barang VALUES (2, 'Kertas HVS', 200, 3);
INSERT INTO persediaan_barang VALUES (3, 'Meja Kantor', 20, 6);
INSERT INTO persediaan_barang VALUES (4, 'Kursi Kantor', 40, 7);
INSERT INTO persediaan_barang VALUES (5, 'Printer', 15, 2);
INSERT INTO persediaan_barang VALUES (6, 'Spidol', 100, 4);
INSERT INTO persediaan_barang VALUES (7, 'Monitor', 30, 8);
INSERT INTO persediaan_barang VALUES (8, 'Papan Tulis', 10, 9);
INSERT INTO persediaan_barang VALUES (9, 'Lemari Arsip', 25, 10);
INSERT INTO persediaan_barang VALUES (10, 'Mouse', 80, 5);

COMMIT;
SET FEEDBACK ON;

SELECT * from department;
SELECT * from employee;
SELECT * from pemasok;
SELECT * from persediaan_barang;

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE pemasok COMPUTE STATISTICS;
ANALYZE TABLE persediaan_barang COMPUTE STATISTICS;

CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;
```

Script Output | Query Result

Task completed in 0.053 seconds

```
Table DEPARTMENT analyzed.

Table EMPLOYEE analyzed.

Table PEMASOK analyzed.

Table PERSEDIAAN_BARANG analyzed.
```

Click on an identifier with the Control key down to perform "Go to Declaration"

Line 125 Column 2 | Insert | Modified | Windows: CR

Oracle SQL Developer : JURNAL_Modul08_Basdat_103122430009

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Connections

Oracle Connections
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Modul08_Basdat_103122430009.sql | JURNAL_Modul08_Basdat_103122430009

Worksheet | Query Builder

```
INSERT INTO persediaan_barang VALUES (3, 'Meja Kantor', 20, 6);
INSERT INTO persediaan_barang VALUES (4, 'Kursi Kantor', 40, 7);
INSERT INTO persediaan_barang VALUES (5, 'Printer', 15, 2);
INSERT INTO persediaan_barang VALUES (6, 'Spidol', 100, 4);
INSERT INTO persediaan_barang VALUES (7, 'Monitor', 30, 8);
INSERT INTO persediaan_barang VALUES (8, 'Papan Tulis', 10, 9);
INSERT INTO persediaan_barang VALUES (9, 'Lemari Arsip', 25, 10);
INSERT INTO persediaan_barang VALUES (10, 'Mouse', 80, 5);

COMMIT;
SET FEEDBACK ON;

SELECT * from department;
SELECT * from employee;
SELECT * from pemasok;
SELECT * from persediaan_barang;

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE pemasok COMPUTE STATISTICS;
ANALYZE TABLE persediaan_barang COMPUTE STATISTICS;

CREATE VIEW empvu80 AS
  SELECT
      employee_id AS id_number,
      last_name AS name,
      salary,
      department_id
  FROM employee
  WHERE department_id = 80;

  DESCRIBE empvu80;
```

Script Output | Query Result

Task completed in 0.072 seconds

```
------------ -------- ------------
ID_NUMBER    NOT NULL NUMBER
NAME         NOT NULL VARCHAR2(50)
SALARY                NUMBER
DEPARTMENT_ID         NUMBER
Name         Null?    Type
------------ -------- ------------
ID_NUMBER    NOT NULL NUMBER
NAME         NOT NULL VARCHAR2(50)
SALARY                NUMBER
DEPARTMENT_ID         NUMBER
```

Click on an identifier with the Control key down to perform "Go to Declaration"

Line 121 Column 7 | Insert | Modified | Windows: C

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Connections

Oracle Connections
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Modul08_Basdat_103122430009.sql | JURNAL_Modul08_Basdat_103122430009

Worksheet | Query Builder

```
INSERT INTO persediaan_barang VALUES (1, 'Laptop', 50, 5);
INSERT INTO persediaan_barang VALUES (2, 'Kertas HVS', 200, 3);
INSERT INTO persediaan_barang VALUES (3, 'Meja Kantor', 20, 6);
INSERT INTO persediaan_barang VALUES (4, 'Kursi Kantor', 40, 7);
INSERT INTO persediaan_barang VALUES (5, 'Printer', 15, 2);
INSERT INTO persediaan_barang VALUES (6, 'Spidol', 100, 4);
INSERT INTO persediaan_barang VALUES (7, 'Monitor', 30, 8);
INSERT INTO persediaan_barang VALUES (8, 'Papan Tulis', 10, 9);
INSERT INTO persediaan_barang VALUES (9, 'Lemari Arsip', 25, 10);
INSERT INTO persediaan_barang VALUES (10, 'Mouse', 80, 5);

COMMIT;
SET FEEDBACK ON;


ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE pemasok COMPUTE STATISTICS;
ANALYZE TABLE persediaan_barang COMPUTE STATISTICS;

CREATE VIEW empvu80 AS
  SELECT
      employee_id AS id_number,
      last_name AS name,
      salary,
      department_id
  FROM employee
  WHERE department_id = 80;

  DESCRIBE empvu80;

  CREATE INDEX idx_emp_dept_id ON employee(department_id);
```

Script Output | Query Result

Task completed in 0.074 seconds

```
View EMPVU80 created.

Name         Null?    Type
------------ -------- ------------
ID_NUMBER    NOT NULL NUMBER
NAME         NOT NULL VARCHAR2(50)
SALARY                NUMBER
DEPARTMENT_ID         NUMBER

Index IDX_EMP_DEPT_ID created.
```

Click on an identifier with the Control key down to perform "Go to Declaration"

Line 120 Column 4 | Insert | Modified | Windows: C